

11 (long)

(a) As shown in Fig. 7(a), a closed ring of soft cast-steel has a coil of insulated wire wound round it a total of N times, carrying a current I . When considering the magnetic induction in the ring, explain whether or not it matters if the winding is uniform round the ring or concentrated over one segment. [4]

(b) Suppose the ring has a cross-sectional area of 1000 mm^2 and a mean circumference of 600 mm . Find the number of ampere-turns required to produce a magnetic flux density of $B = 1 \text{ T}$ inside the ring. The relevant B - H curve for cast-steel is given in Fig. 8. [7]

(c) A cut is made through the ring leaving a gap of 2 mm as shown in Fig. 7(b). Find the number of ampere-turns required to produce the magnetic flux density of $B = 1 \text{ T}$. Explain the difference between this answer and that in part (b) above. [8]

(d) The effect of the magnetic field is to exert forces that would tend to close the gap. Estimate the force that would need to be applied to keep the gap at 2 mm , making explicit any approximations made. [8]

(e) How is the gap maintained in practice? [3]

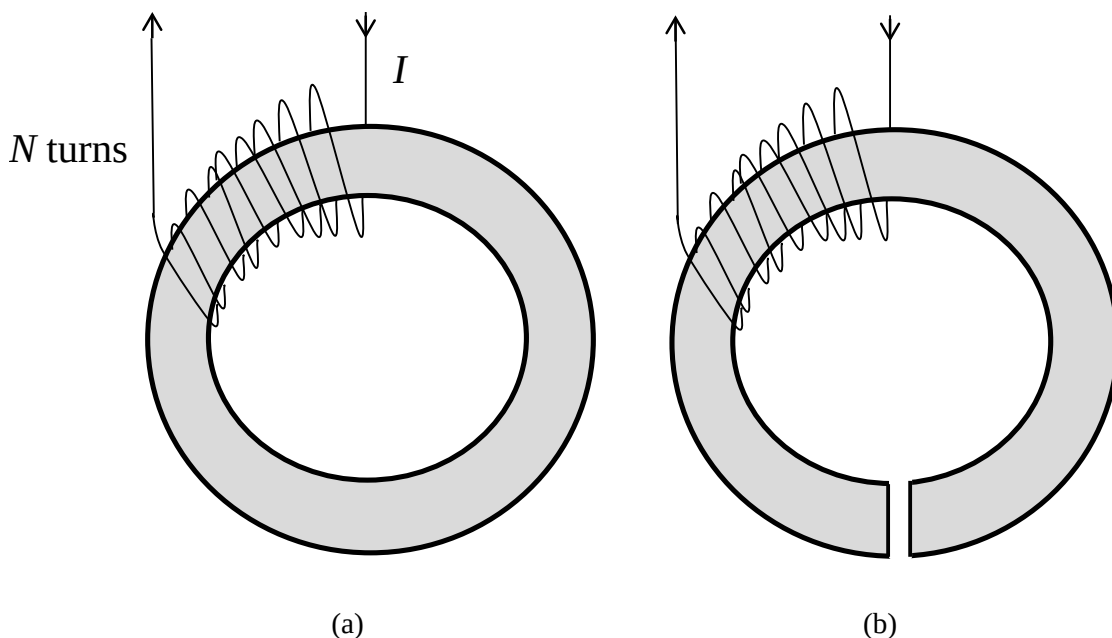


Fig. 7

cont.

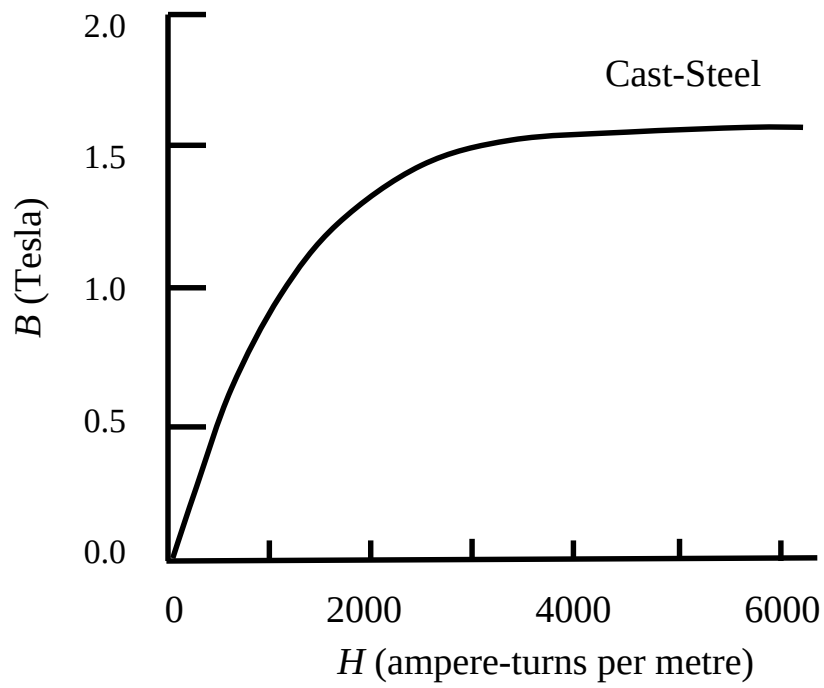


Fig. 8